Model-Theoretic Signature $\Sigma = (P; 1; S, +, \cdot, \text{^})$

SORT P — the carrier set (the Peano system)

One sort only. All function symbols have domain and codomain in P.

CONSTANT $1 : P$ — the distinguished initial element

Axiom: $1 \notin \mathrm{rng}(S)$ (1 is not a successor)

FUNCTION SYMBOLS

Symbol	Type	Arity	Base clause	Successor clause
S	$P \to P$	1	—	$S(n) \neq S(m) \to n \neq m$ (injective)
+	$P \times P \to P$	2	$m + 1 = S(m)$	$m + S(n) = S(m + n)$
·	$P \times P \to P$	2	$m \cdot 1 = m$	$m \cdot S(n) = (m \cdot n) + m$
^	$P \times P \to P$	2	$m \text{ ^ } 1 = m$	$m \text{ ^ } S(n) = (m \text{ ^ } n) \cdot m$

Dependency: S (primitive) → + → · → ^ (each step rule uses the previous operation)

PEANO AXIOMS (for S and 1)

P1: $1 \in P$

P2: $S : P \to P$

P3: $1 \notin \mathrm{rng}(S)$

P4: S injective

P5: Induction (second-order)

ALGEBRAIC LAWS (to be proved)

Commutativity: $m + n = n + m$

Associativity: $(m+n)+k = m+(n+k)$

Distributivity: $m \cdot (n+k) = m \cdot n + m \cdot k$

Cancellation: $m+k=n+k \to m=n$

Exp laws: $m\text{^}(n+k) = m\text{^}n \cdot m\text{^}k$

Categoricity: any two Σ-structures satisfying P1–P5 are uniquely isomorphic (Uniqueness of Peano Systems up to Isomorphism)